Zhekai Jiang

Doctoral Assistant
Data Analysis Theory and
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EPFL

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Education

Docteur ès sciences (Ph.D.) in computer science — Sep 2024–Present
EPFL — Lausanne, Switzerland

- Thesis advisor: Prof. Christoph Koch
- Tentative thesis title: Structural Decomposition Algorithms for Database Query Optimization
- Cumulative Grade Point Average: 6.00 / 6.00
- Course highlights: Systems for Data Management and Data Science, Formal Verification

Bachelor of Software Engineering — Sep 2019–May 2024
McGill University — Montréal, Canada

- Cumulative Grade Point Average: 4.00 / 4.00
- Course highlights: Database Systems, Compiler Design, Model-Based Programming, Programming Languages and Paradigms, Software Language Engineering, Applied Machine Learning, Theory of Computation, Algorithms and Data Structures, Oral and Written French (B2 level), Physics of Music

Work Experience

Research Intern — Jan 2023–Aug 2023
Oracle Labs — Zürich, Switzerland

- Developed a scalable analytics dashboard (Java/Dropwizard, TypeScript/Oracle JET) to interactively visualize millions of historical download records over several years, with zoomable charts and dynamic filtering
- Designed REST APIs and optimized SQL queries (JDBI) to support complex, high-volume data requests, deployed as Docker containers on the Oracle Cloud Infrastructure (OCI)
- Implemented and optimized automated testing in CI/CD that sped up testing by 10x+
- Led the full development lifecycle, from translating stakeholder requirements into technical specifications to configuring CI/CD pipelines and automated testing
- Presented the overall project architecture and features on behalf of the team in an invited internal tech talk

Research Intern — May 2022–Aug 2022
EPFL — Lausanne, Switzerland

- Developed data aggregation solvers (Scala) in the project Sudokube, a high-performance online analytical processing (OLAP) database system for data analysis based on data cubes
- Developed a web interface (TypeScript/React.js, gRPC) to allow users to execute queries and visualize results
- Demonstrated the system at the 2023 International Conference on Management of Data (SIGMOD '23)

Teaching Experience

Teaching Assistant, Making Intelligent Things (EPFL CS-358 – Spring 2025, Fall 2025, Spring 2026)

- Advise students on individual and group projects prototyping intelligent physical artifacts, combining microcontroller programming, electronics, and computer-aided design and manufacturing, following agile workflow
- Grade quizzes and project deliverables

Mentor, Linear Algebra and Geometry (McGill MATH 133 – Fall 2020, Fall 2021, Fall 2022)

- Hosted weekly office hours and monitored the online discussion forum
- Wrote about applications of linear algebra to other areas of mathematics and computer science, as extra reading for the interested

Mentor, Programming Language and Paradigms (McGill COMP 302 – Winter 2022)

- Hosted weekly office hours

Mentor / Grader, Introduction to Software Engineering (McGill ECSE 321 – Winter 2021, Fall 2021)
- Advised students on group project (a full-stack software application including a backend server, a frontend website, and a mobile application, following agile workflow)
- Graded exams with design problems on requirements engineering and data modelling

Awards and Honours

- EPFL School of Computer and Communication Sciences PhD Fellowship — Sep 2024–Aug 2025
- McGill Charles Michael Morssen Gold Medal — May 2024
- McGill Dean's Honour List — Jul 2020, Jul 2021, Jul 2022, Sep 2023, May 2024
- McGill Tomlinson Engagement Award for Mentoring — Dec 2020, Apr 2021, Dec 2021, Apr 2022, Dec 2022
- McGill Hatch Scholarship in Engineering — Jul 2022
- EPFL Scholarship of Excellence — May 2022–Aug 2022
- McGill Schull–Yang International Experience Award / Summer Undergraduate Research Award in Engineering — May 2022
- McGill Engineering Class of 1983 Scholarship — Jul 2021
- McGill Rio Tinto–Richard Evans International Exchange Award — Jul 2020
- McGill John V. Galley Scholarship — Jul 2020
- HKU Award of Excellence — Sep 2018

Publications

(* indicates equal contribution; † indicates alphabetical author ordering, as is conventional in theoretical venues.)

Full Papers

[f1] **Bound-Adorned Datalog**
Christian Fattebert†, Zhekai Jiang†, Christoph Koch†, Reinhard Pichler†, and Qichen Wang†
PODS '26: *45th Symposium on Principles of Database Systems, Bengaluru, India, May 2026*

[f2] **Succinct Structure Representations for Efficient Query Optimization**
Zhekai Jiang, Qichen Wang, and Christoph Koch
SIGMOD '26: *International Conference on Management of Data, Bengaluru, India, May 2026*

Demo Papers

[d1] **Concretize: A Model-Driven Tool for Scenario-Based Autonomous Vehicle Testing**
Jerry Hou-Liu*, Zhekai Jiang*, and Aren A. Babikian
MODELS '24: *ACM/IEEE 27th International Conference on Model Driven Engineering Languages and Systems, Linz, Austria, Sep 2024*

[d2] **Aggregation and Exploration of High-Dimensional Data Using the Sudokube Data Cube Engine**
Sachin Basil John, Peter Lindner, Zhekai Jiang, and Christoph Koch
SIGMOD '23: *International Conference on Management of Data, Seattle, USA, Jun 2023*

Short Papers

[s1] **Towards a Traffic Scenario Catalog for Collaborative Testing of Autonomous Vehicles**
Zhekai Jiang, Oszkár Semeráth, and Aren A. Babikian
SE4ADS '25: *IEEE/ACM 1st International Workshop on Software Engineering for Autonomous Driving Systems, Ottawa, Canada, Apr 2025*

[s2] **OptObstacles at the SBFT 2025 Tool Competition – UAV Testing Track**
Zhekai Jiang and Aren A. Babikian
SBFT '25: *IEEE/ACM International Workshop on Search-Based and Fuzz Testing, Ottawa, Canada, Apr 2025*

Service

McGill International Experience Awards Embassador — Sep 2022–Present
- Shared my international experience with the McGill community and perspective research interns of EPFL